MILITARY SPECIFICATION

GENERATING AND CHARGING PLANT, 5 TON PER DAY CAPACITY, SKID MOUNTED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercial fixed type generating and charging plant designed for the production of oxygen and nitrogen.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	Plastic Sheet and Strip, Thin Guage, Polyolefin.
P-C-436	Cleaning Compound, Alkali, Boiling Vat (Soak) or Hydrosteam.
P-C-437	Cleaning Compound, High Pressure (Steam) Cleaner.
BB-N-411	Nitrogen, Technical.
CC-M-1807	Motor, Alternating-Current, Fractional and Integral Horsepower (500 HP and Smaller).
TT-P-664	Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-V-121	Varnish, Spar, Water-Resisting.
UU-T-81	Tags, Shipping and Stock.
PPP-P-40	Packaging and Packing of Hand Tools.
PPP-B-566	Boxes, Folding, Paperboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB TX 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3655

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	PPP-B-601	Boxes, Wood, Cleated-Plywood.
	PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
	PPP-B-636	Box, Shipping, Fiberboard.
	PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall.
	PPP-B-665	Boxes, Paperboard, Metal Edged and Components.
	PPP-B-676	Boxes, Setup.
	PPP-C-843	Cushioning Material, Cellulosic.
	PPP-T-60	Tape, Packaging, Waterproof.
	PPP-T-97	Tape, Pressure Sensitive, Filament Reinforced.

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	MIL-V-3	Valves, Fittings, and Flanges (except for Systems Indicated Herein); Packaging of.
	MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
*	MIL-P-116	Preservation, Methods of.
	MIL-B-117	Bags, Sleeves and Tubing-Interior Packaging.
	MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible.
	MIL-P-130	Paper, Wrapping, Laminated and Creped.
*	MIL-B-131	Barrier Materials, Water Vaporproof, Greaseproof, Flexible, Heat-Sealable.
*	MIL-V-173	Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
	DOD-D-1000	Drawings, Engineering and Associated Lists.
*	MIL-R-3065	Rubber, Fabricated Products.
	MIL-R-3593	Refrigeration and Cooling Equipment (excluding Household Refrigerators), Packaging of.
	MIL-C-3600	Compressor, Rotary, Power-Drive; and Compressors, Reciprocating, Power-Driven: Air and Gas (except Oxygen and Refrigerant), Packaging of.
	MIL-C-3774	Crates, Wood; Open 12,000 and 16,000 Pound Capacity.
*	MIL-C-5501	Caps and Plugs; Protective, Dust and Moisture Seal.
	MIL-C-9897	Crates, Slotted Angle, Steel or Aluminum, for Lightweight Airframe Components and Bulky Item (for Maximum Loads of 3,000 Pounds).
	MIL-C-15074	Corrosion Preventive, Fingerprint Remover.
	MIL-D-16791	Detergents, General Purpose (Liquid, Nonionic).
	MIL-P-17667	Paper, Wrapping, Chemically Neutral (Non-Corrosive).
*	MIL-C-23760	Coating, Sprayable, Strippable, Protective, for Packaging of Weapons Systems and Components; Application of.
	MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas.
	MIL-H-27301	Hose Assembly, Metal, Cryogenic Liquid Transfer.
*	MIL-T-27730	Tape, Antiseize, Polytetrafluoroethylene, with Dispenser.
	MIL-C-52950	Crates, Wood, Open and Covered.
	MIL-C-81302	Cleaning Compound, Solvent, Trichlorotrifluoroethane.

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STANDARDS

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MIL-STD-101	Color Code for Pipelines and for Compressed-Gas Cylinders.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-163	Steel Mill Products Preparation for Shipment and Storage.
MIL-STD-649	Aluminum and Magnesium Products Preparation for Shipment and Storage.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting offcer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd, Chicago, IL 60606.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Code (Section VIII, Unfired Pressure Vessels).

(Copies of ASME publications may be obtained from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the test of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) a sample shall be subject to first article inspection (see 4.4).

3.2 Components. The generating and charging plant shall be a complete self contained electrical power driven unit consisting of all components needed for compliance with the requirements specified herein. Each component shall include all the parts, controls, instruments, and accessories necessary for its safe and effective operation.

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3.3 Materials. Materials shall be as specified herein. Material not definitely specified shall be of the quality regularly used by the manufacturer for sale to the commercial market. Materials shall be free from all defects and imperfections that will affect the serviceability and appearance of the finished product.

3.3.1 Fungus proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with varnish conforming to MIL-V-173. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.2 Metals. All metallic surfaces of the generating and charging plant that contact high purity oxygen or nitrogen and that are in a location where they will normally be chilled to cause condensation of moisture from the atmosphere, shall be fabricated from corrosion resistant materials that do not require paint or other organic chemical coatings for protection.

* 3.3.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.3 Gasketing and insulating materials. Plastic, rubber, or similar gasketing and insulating materials shall be compounded to insure their suitability for the intended application and, where applicable, their resistance to hydrocarbons or low temperatures. Packings, gaskets, seals, and other nonmetallic components that come into contact with high purity oxygen shall be compounded of materials compatible with use in a pure oxygen atmosphere, under conditions to which they will be subjected, and that are not adversely affected or deteriorated by continued use in an oxygen or nitrogen atmosphere.

3.3.4 Rubber goods. Rubber goods such as hoses, seals, insulation or isolation boots, et cetera, shall be fabricated from rubber compounds conforming to MIL-R-3065.

3.3.5 Flammable materials. Readily flammable materials shall not be used where they come in contact with high purity oxygen.

3.3.6 Pipe thread tape. Pipe thread tape conforming to MIL-T-27730 shall be applied to threaded fittings subject to contact with liquid oxygen.

3.4 Design. The generating and charging plant shall be a standard, heavy duty unit that can be erected, operated and maintained in a building with a previously prepared concrete foundation. Heavy duty commercially proven generating and charging equipment components shall be used to the maximum practicable extent. All electric switch gear, wiring and lighting equipment (exclusive of building lighting facilities), piping, ducting, instruments and accessory components needed for installation, operation and maintenance of the generating plant shall be provided.

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3.4.1 Tiedown and lifting provisions.

3.4.1.1 Tiedowns. The generating and charging plant components furnished shall not require special mounting provisions other than tie down bolts for installation on a concrete foundation.

3.4.1.2 Lifting rings. Each shipping section shall be provided with lifting rings in locations that will permit cable attachment and lifting without damage to any component.

3.4.1.3 Lifting eyes. Lifting eyes or other hoisting attachments shall be provided on major components so that they may be readily removed or replaced. Insofar as practicable, all individual components shall be installed in a manner that will permit their removal without necessity for removing other components.

* 3.4.2 Electric Motors. The electric motor(s) furnished shall be adequate for the application. The electrical characteristics for the motors shall be 230/460 volts, 3 phase, 50/60 cycle, totally enclosed type, equipped with ball bearings and shall conform to specification CC-M-1807, Type II, Class 3, Style E, Design K.

3.4.2.1 Starter and Overload Protection. Each electric motor rated at 30 horsepower or greater shall be equipped with an auto-transformer type reduced voltage starter. The starters shall include adequate under voltage and over voltage protection.

3.4.2.2 Voltage Conversion. Voltage conversion from 230 to 460 volts and vice versa shall be accomplished by changing control transformer taps, starter heater elements and motor winding connections. When delivered, the plant shall be connected for 460 volts operation. A reversible warning plate shall be permanently attached adjacent to the transformer indicating the voltage (230 or 460 V) at which the unit is to be operated as well as the applicable connections.

3.4.2.3 Motor Starting Switches. Each of the electric motors in the plant shall be equipped with a magnetic starting switch of the proper size and type. The magnetic switch shall provide under voltage and over current protection and shall be of the manual reset type suitable for the use intended. All magnetic starters shall be equipped with 110 volt AC actuating coils.

3.4.2.4 Control Voltage Transformer. The voltage employed for the magnetic control of the motors shall be 100 volts/50/60 cycle/1 phase. A step down transformer of capacity as required shall provide the control voltage.

3.4.2.5 Junction Box. The plant shall be equipped with a main junction box which shall include the motor magnetic switches. The junction box shall be completely weatherproof and a hinged cover shall be provided. Means shall be provided to allow resetting of the magnetic switches without opening the cover.

3.4.2.6 Circuit Breaker. The electric motors shall be protected from over current by means of a circuit breaker. The circuit breaker shall be of the type which is nonsensitive to temperature and shall be capable of tripping out the electric motors during overloading conditions and shall be equipped with manual reset.

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3.4.3 Carbon dioxide removal. Carbon dioxide will be removed from the air process stream. A standard commercial proven process, other than caustic scrubbing will be used for the carbon dioxide removal.

3.4.4 Installation.

* 3.4.4.1 Components within building. The generator and charging plant design shall provide for installation of the electric motors, air compressor, refrigeration unit, air purification equipment, air processing equipment, rectification equipment, instruments, controls and other critical components within the building area. Complete foundation specifications, equipment layout, installation drawings and special foundation bolts shall be furnished with each generating plant.

3.4.4.2 Components external to building. All components that require external installation to increase operating efficiency shall be installed external to the building and shall operate under the conditions specified herein without other than self contained environmental protection.

3.4.5 Supplies. The generating and charging plant shall be designed to perform as specified herein when operated with electric power having the characteristics specified in 3.4.10. It shall require no operating supplies other than electric power, lubricants, water, antifreeze, chemicals for purity test and repair and maintenance tools and materials.

3.4.6 Life. The generating and charging plant components shall be designed for a minimum operational life of 10,000 hours between overhauls and to minimize loss of production time for maintenance, repair and periodic servicing.

3.4.7 Lubrication. Wherever practical, permanently lubricated components, or components not requiring lubrication shall be used in the design of the generating and charging plant. Grease and oil seals shall be designed and located to provide maximum accessibility for inspection and replacement.

3.4.8 Parts subject to wear. The generating and charging plant shall be designed to provide for lubrication of all parts subject to wear. Wherever practical, such parts shall be designed and constructed to be readily renewable. All grease fittings and other lubrication points shall be located to be readily accessible.

3.4.9 Detachable fittings. Wherever practicable, the generating and charging plant shall be designed to include disconnect plugs, receptacles, and multiline connectors in the electrical system and readily detachable and attachable type fittings in liquid and vapor lines to permit rapid component removal and replacement. All disconnect points shall be clearly indicated and identified.

* 3.4.10 Electrical Equipment. All electrical equipment shall be constructed and properly installed to insure satisfactory performance in accordance with the requirements specified herein. When the generating and charging plant is furnished with 230/460 volts /3 phase/ 50/60 cycles, all electrical equipment shall be capable of operating at full load at either frequency without overheating or damage. All switches, circuit breakers and motor starting devices needed for safe and effective plant operation shall be provided.

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3.4.11 Controls. Insofar as practical, all manually operated controls used in normal operation of the generating and charging plant shall be grouped into control panels. They shall be located in positions readily visible and accessible to an operator standing at the control panel.

3.4.12 Instruments. All instruments necessary for indicating the information needed in controlling the equipment and determining its performance shall be grouped into control or instrument panels and securely mounted thereon in positions readily visible and accessible to operation and maintenance personnel.

3.4.13 Pressure gages. Pressure gages shall be equipped with pulsation snubbers and those subject to vibration shall be mounted in rubber or placed on a panel isolated from the vibration source.

3.4.14 Pressure relief devices. Pressure relief devices shall be provided in any line or component that can be so isolated by closing valves, or otherwise, where dangerous pressures could develop. Relief devices shall be provided for normal relief of pressure. All pressure relief devices shall be installed in readily accessible location. Provisions shall be incorporated to insure discharge of pressure without damage to equipment or danger to personnel. Safety valves shall be sized in accordance with the ANSI/ ASME BVP VII - Boiler and Pressure Vessel Code and selected from the list published in ASME PVP Vol. 33, Safety Relief Valves approved by the National Board of Boiler and Pressure Vessel Inspector.

3.4.15 Blow-out type safety heads. Blow-out type safety heads shall be provided where pressure will not normally have to be relieved, but where protection is necessary. Examples are air processing and separation equipment insulated casings.

3.4.16 Safety devices. The plants shall be equipped with audible warning alarms or shut down devices that actuate when unusual operating conditions occur that could cause plant damage or endanger personnel if operation were allowed to continue.

3.5 Performance.

3.5.1 Production capacity. When operating at normal atmospheric pressure, in an ambient temperature (outside enclosed building) of 70° to 100° Fahrenheit (F) in a relative humidity of 100 percent, the generating and charging plant shall be capable of delivering liquid product through the plant servicing lines and gaseous product through the charging manifold assembly at a daily average of not less than the rates specified in Table I.

TABLE I. Delivery Capabilities

	Oxygen Liquid	Oxygen Gas	Nitrogen Liquid	Nitrogen Gas
Production Capacity	417 LB/HR	5000 SCFH*	250 LB/HR	3400 SCFH*
Delivery Pressure Pounds Per Square Inch Gage (Psig)	5 MINIMUM	0-3500	5 MINIMUM	0-3500

* SCFH - Standard (760 millimeters of mercury, 70°F) cubic feet per hour.

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3.5.2 Concurrent product delivery. During concurrent production, the generating and charging plant shall be capable of delivering each product, the delivery rate of each being reduced from the rates in 3.5.1 by not more than the percentage of full capacity at which the other is delivered.

3.5.2.1 The generating and charging plant shall be capable of concurrently delivering liquid and gaseous product through the lines specified in 3.5.1 as follows:

a. Liquid oxygen 5 psig (minimum) and gaseous nitrogen 3500 psig.

b. Liquid nitrogen 5 psig (minimum) and gaseous oxygen 3500 psig.

c. Liquid nitrogen 5 psig (minimum) and liquid oxygen 5 psig (minimum).

3.5.3 Product quality.

3.5.3.1 Liquid oxygen. The liquid oxygen product shall conform to type II requirements specified in MIL-O-27210.

3.5.3.2 Gaseous oxygen. The gaseous oxygen product shall conform to the type 1 requirements specified in MIL-O-27210.

3.5.3.3 Liquid nitrogen. The liquid nitrogen product shall conform to type II, class 1, grade B requirements specified in BB-N-411.

3.5.3.4 Gaseous nitrogen. The gaseous nitrogen product shall conform to type I, class 1, grade B requirements specified in BB-N-411.

* 3.5.4 Air Compression Equipment. A multi-stage air compressor with a minimum of five stages at 3:1 ratio per stage shall be provided and shall furnish all the compressed air required in the separation cycle of the generating and charging plant. The air compression equipment group, which consists of inter and aftercoolers, gauge panel, safety relief devices etc, shall be a complete self contained assembly requiring only electric power supply, lubricants, and cooling system fluid for operation. A condensate blow down valve shall be provided for each stage of the air compressor and shall be equal to the type manufactured by Barksdale Control Division, Transamerica De Lavel.

3.5.4.1 Air compression equipment shall be commercially proven, heavy-duty, low speed and designed for continuous operation when subjected to ambient conditions outlined herein.

3.5.5 Liquid product delivery provisions. The generating and charging plant air processing and separation equipment shall include separate liquid product delivery provisions for each of the two liquid products.

3.5.6 Liquid servicing. The liquid product, or oxygen/nitrogen, servicing coupling of the generating and charging plant shall be designed to be compatible with hose assembly, metal, cryogenic liquid, transfer, 3/8 inch inside diameter, MIL-H-27301.

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3.5.7 Gaseous product servicing system. The gaseous product servicing system shall consist of a gaseous product delivery line, a charging manifold assembly, and the charging leads, and other accessories necessary for filling high-pressure storage containers from the charging manifold.

3.5.7.1 Charging manifold. The charging manifold shall consist of two independent charging sections, plus mountings, controls, instruments and accessories needed for safe and efficient operation.

3.5.7.2 Manifold sections. Each manifold section shall include a minimum of twenty (20) charging stations. The height of the stations and its fittings shall be compatible with the valve connections on the standard fifty-one inch oxygen or nitrogen gas storage cylinders. Each individual charging station shall be equipped with a manually operated shut off valve capable of handling 3,500 pounds per square inch (psi) pure oxygen/nitrogen gas.

3.5.7.3 Cylinder Evacuation. Each oxygen manifold section shall have provisions for evacuating twenty (20) standard fifty-one inch oxygen cylinders to an absolute pressure of three inches of mercury in less than fifteen minutes and maintaining this pressure until cylinder filling begins. One vacuum pump capable of meeting this requirement shall be furnished with each generating and charging plant.

3.5.8 Environmental characteristics.

3.5.8.1 Operating temperature. The generating and charging plant shall be capable of starting and operating continuously at normal (not necessarily rated) production when the ambient temperature (outside the enclosed building) between $-65^{\circ}F$ and $+125^{\circ}F$, including exposure to solar radiation of 100 to 120 watts per square foot at the higher temperature.

3.5.8.2 Humidity. The generating and charging plant shall be capable of operating at temperatures referenced in 3.5.1 and 3.5.8.1 and in a relative humidity of 100 percent.

3.5.8.3 Salt-sea atmosphere. The generating and charging plant shall not be damaged by operation or storage in an atmosphere containing salt-laden moisture such as encountered near bodies of salt water and in transportation on shipboard.

3.6 Maintainability.

3.6.1 Components and accessories. Components and accessories shall be constructed and installed on each individual skid-mounted shipping section to provide for:

a. Ready accessibility to all parts for operation, maintenance, and repairs in the field.

b. Replacement and adjustment of components and accessories with minimum disturbance to other elements of the equipment.

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c. Covers or plates that must be removed to accomplish normal operation and maintenance procedures or for frequent component or part removal shall be equipped with substantial quick-disconnect fittings.

3.6.2 Operation and maintenance tools. A complete set of special tools (other than those common hand tools normally available in Wing/Base maintenance shops) required for proper operation and maintenance of the generating and charging plant components shall be provided.

* 3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.8 Diagrams and instructions.

3.8.1 Flow diagram. A flow diagram illustrating the generating and charging plant overall processes and equipment shall be permanently affixed on or near the air processing and separation equipment control panel in a readily accessible and protected location. The flow diagram shall be of sufficient size for clear readability, and shall be made from sheet aluminum or sheet zinc of not less than 0.050 inch thickness, anodized, or etched to produce raised markings with a black or other dark color background, and with a raised border of not less than 1/2 inch width.

3.8.2 Electrical wiring diagram. A complete electrical wiring diagram detailing the generating and charging plant electrical systems shall be permanently affixed on or near the air processing and separation equipment control panel in a readily accessible and protected location. The wiring diagram shall be of sufficient size for clear readability, and shall be made of sheet aluminum or sheet zinc as specified in 3.8.1.

3.9 Color identification system. Exposed air source and product piping shall be marked in accordance with MIL-STD-101.

3.10 Finishes and protective coatings. Other than color coding as prescribed in 3.9, all surfaces shall be finished in accordance with standard commercial practices.

3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12 Workmanship. All generating and charging plant parts shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from defects, burrs, and sharp edges, radii or fillets, thoroughness of welding, brazing, soldering, painting, wiring, alignment of parts, tightness and assembly of bolts, screws, et cetera.

3.12.1 Cleaning. Components of the generating and charging plant shall be thoroughly cleaned to remove dirt, grease, excess soldering, brazing and welding flux, welding slag, loose, spattered or excess solder, metal chips, and other foreign materials before, during, and after assembly.

* 3.13 Reclaimed materiels. The use of reclaimed materiels shall be encouraged to the maximum extent possible.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspections. Inspection and testing of the generating and charging plan shall be classified as follows:

a. First Article Inspection ..... see 4.4.

b. Quality Conformance Inspection ..... see 4.5.

4.3 Test conditions.

4.3.1 Preparation for test. The generating and charging plant shall be erected and run-in as the manufacturer determines necessary for his preliminary check-out operation. All necessary adjustments other than normal control adjustments shall be made during the run-in and shall remain undisturbed thereafter. Following the running-in, the generating and charging plant shall be defrosted and serviced for normal operation.

4.3.2 Operating conditions. Operation throughout the first article and quality conformance inspection shall be continuous except that the generating and charging plant may be shut down for no longer than one (1) hour out of each twenty-four (24) hours (1 hour each calendar day) if necessary for lubrication, minor adjustments, repairs, inspections and preventive maintenance without penalty.

4.3.2.1 If an interruption of a greater length than allowed in 4.3.2 (totaling more than one (1) hour in a twenty-four (24) hour period) is necessary and results in a production break of twelve (12) hours or less, the required production time shall be extended by two (2) hours for each hour of shutdown time plus the operating time necessary for resuming the production rates specified in Table I. Example: A five (5) hour shutdown requiring an additional three (3) hours of operation to resume rated production would extend the test period by thirteen (13) hours of actual production in addition to that specified herein. In case of necessity for shutdown in excess of the permissible hour, this one (1) hour shall not be omitted from the shutdown time total when computing the specified extension as required production time. If a production break of more than twelve (12) hours is necessary, the purpose of the test shall be considered defeated and the test shall be repeated unless continuation is authorized by the procuring activity.

4.3.3 Apparatus. Insofar as practicable, apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at intervals properly spaced to insure continued laboratory accuracy.

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4.3.4 Tools. Special tools and field equipment shall be used to the maximum practicable extent during testing. Utilization of these tools shall be sufficient to determine their usefulness. Instances where the special tools furnished are inadequate shall be reported in detail.

4.3.5 Pressures. Pressures and pressure differentials shall be measured by means of the pressure gauges and differential pressure meters furnished as part of the generating and charging plant.

4.3.6 Temperatures. Temperatures shall be recorded in degrees Fahrenheit.

4.3.7 Production rate.

* 4.3.7.1 Liquid Product. Liquid oxygen and liquid nitrogen production rates shall be determined by one of the following methods:

* a. Weighing the liquid delivered into a container maintained at atmospheric pressure.

b. Using a cryogenic liquid flowmeter accurate to 0.5 percent certified by or traceable to, The National Bureau of Standards.

c. Vaporizing the liquid product into a totaling type gas flowmeter accurate to 0.5 percent as certified by, or traceable to, The National Bureau of Standards.

In addition to methods b. and c. above, a minimum of 12 hours weighing of liquid product shall be mandatory.

* 4.3.7.1.1 Collection Container Calibration. Containers for collection and weighing of liquid product shall be calibrated for heat leak evaporation losses. The weight of liquid found to evaporate from the container in a length of time equal to a production collection period may be added to the weight collected to determine the true product delivery capacity. Heat leak evaporation loss rate determinations shall be made only when the container is not less than one-half full, and when the filled container has sat for not less than 12 hours after filling to permit stabilization of temperatures in the container and its insulation. Allowance shall not be permitted for liquid evaporated in the transfer hose connecting the container to the generating and charging plant.

* 4.3.7.2 Gaseous Products. Gaseous oxygen and gaseous nitrogen production rates shall be determined by the increase of pressure in high pressure gas storage cylinders being charged to not less than 3,500 psig or by passing the vapors through the gaseous product servicing system specified in 3.5.7 at not less than 3,500 psig, expanding the vapors to gas meter inlet pressure, and measuring the gas volume with a properly calibrated recording or totaling type gas flowmeter designed to handle the product.

4.3.7.2.1 Gaseous oxygen and gaseous nitrogen charging rates shall be determined by passing the vapors through the charging manifold assembly at the specified pressure, expanding the vapors to gas meter inlet pressure after the manifold, and measuring the gas volume with a properly calibrated recording or totaling type gas flowmeter designed to handle the product.

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4.3.7.3 Column control. Throughout the production capacity measurement period, the column liquid levels and generating and charging plant pressures shall be maintained within the limits recommended by the manufacturer.

4.3.8 Observations. During the progress of all tests, such characteristics as smoothness of operation, vibrations, ease of control, speed regulation, pressure regulation, freedom from leaks and the operation of auxiliary equipment shall be observed to determine whether the generating and charging plant complies with all requirements specified herein.

4.3.9 Test data. The data shall be recorded at intervals not to exceed one (1) hour. All operational data will be recorded that has an affect on the operational capabilities of the plant.

4.4 First Article Testing.

4.4.1 Test sample. The test sample shall consist of one generating and charging plant representative of the production equipment. The sample shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Test report. Upon completion of first article inspection, the contractor shall submit a test report prepared in accordance with MIL-STD-831.

4.4.3 First Article Tests. The first article tests shall consist of all tests specified in 4.6 except the acceptance performance run specified in 4.6.6.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of individual tests.

4.5.1 Individual tests. Each generating and charging plant shall be subjected to the following tests as described under "test methods" in 4.6.

a. Examination of product ..... see 4.6.1.

b. Mechanical inspection ..... see 4.6.2.

c. Functional check ..... see 4.6.3.

d. Pressure test ..... see 4.6.4.

e. Acceptance performance run ..... see 4.6.6.

4.6 Test methods.

4.6.1 Examination of product. The overall generating and charging plant and its individual components shall be examined for compliance with the requirements specified herein.

4.6.2 Mechanical inspection. A mechanical inspection of all components and parts of the generating and charging plant for defects, damage and completeness shall be conducted.

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4.6.3 Functional check. All mechanical parts of the generating and charging plant shall be checked for functioning and shall operate freely and properly.

4.6.4 Pressure test. Each component, accessory and circuit of the generating and charging plant that operates under positive pressure in normal service shall be pressure tested in accordance with the ASME Unfired Pressure Vessel Code. Automatic pressure relief valves or blow-out type safety heads need not be installed for this testing. Distortion, damage, or leaks resulting from the pressure testing shall be cause for rejection of the components tested.

4.6.5 First Article performance run. The generating and charging plant shall be prepared for test, as specified in 4.3.1, and production runs accomplished for the time periods shown below. The production rates shall be as specified in 3.5.1 and 3.5.2.

*	Liquid oxygen	300 hours
*	Liquid nitrogen	100 hours
	Gaseous oxygen	100 hours
	Gaseous nitrogen	100 hours
	Concurrent liquid oxygen/nitrogen	100 hours

4.6.5.1 The First Article Inspection will not be considered complete until every major part of component of the plant has passed the test to the satisfaction of the government representative.

4.6.6 Acceptance performance run. The generating and charging plant shall be prepared for test, as specified in 4.3.1. Production runs shall be as follows with rates, as specified in 3.5.1 and 3.5.2.

Liquid oxygen	25 hours
Liquid nitrogen	25 hours
Gaseous oxygen	25 hours
Gaseous nitrogen	25 hours
Concurrent liquid oxygen/nitrogen	25 hours

4.7 Inspection of the preservation, packaging and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PACKAGING

5.1 Cleaning, preservation and packaging. Cleaning, preservation, and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The generating and charging plant, parts, fittings, components and assemblies that come in direct contact with high-purity liquid or gaseous oxygen during operation shall be thoroughly cleaned and dried, by one, or a combination of the following methods:

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5.1.1.1.1. Method a. Vapor degreasing process followed by fingerprint removal. Use trichlorotrifluoroethane conforming to MIL-C-81302, type I, in a standard commercial vapor degreaser. The operation of the degreaser shall be in accordance with the manufacturer's recommendations. Fingerprint remover shall conform to MIL-C-15074. Following application of the fingerprint remover, surfaces shall be thoroughly rinsed with clean trichloroethylene. All solvents shall be removed after the degreasing operation by purging with hot filtered oil-free air or nitrogen, or by vacuum evacuation.

5.1.1.1.2 Method b. Solvent degreasing by slushing methods. All surfaces requiring decreasing shall be thoroughly washed with solvent conforming *to MIL-C-81302. Degreased surfaces shall be cleaned with fingerprint remover conforming to MIL-C-15074. Following application of the fingerprint remover, the surfaces shall be thoroughly rinsed with clean trichloroethylene. All solvents shall be removed after the degreasing operation by purging with hot filtered oil-free air or nitrogen, or by vacuum evacuation. Precautions shall be taken to insure the chlorinated solvent does not contact plastics or other noncompatible materials.

5.1.1.1.3 Method c. Alkali detergent degreasing. Components shall be flushed with hot alkaline cleaning compound conforming to P-C-436, until free from oil, grease, and other combustible materials. Following alkaline cleaning, all surfaces, internal and external, as applicable, shall be thoroughly rinsed with clean hot water and dried by purging with hot, filtered oil-free air or nitrogen or by baking at a temperature of 250° to 300°F, until all water is removed.

5.1.1.1.4 Method d. Steam cleaning. Cleaning shall be effected by subjecting the component to a stream of steam only or to a stream of steam with an added cleaning compound followed by steam alone. Cleaning compound shall conform to P-C-437. Following steam cleaning, all surfaces, internal, and external as applicable, shall be purged with hot filtered oil-free air or nitrogen or by baking at a temperature of 250° to 300°F, until all water is removed.

5.1.1.1.5 Method e. Detergent cleaning. This procedure is applicable to plastics, O-rings, packing, and similar materials. Components shall be cleaned with 4 percent detergent solution (mix 6 ounces of detergent with one gallon of water), for 30 minutes at 120°F. Detergent shall conform to MIL-D-16791. After cleaning, rinse with distilled water several times. Dry by purging with hot filtered oil-free air or nitrogen.

5.1.1.1.6 Method f. Ultrasonic cleaning. Many cleaning solutions can be used as an ultrasonic cleaning media. These liquids may be either water base detergents or organic solvents. Care shall be taken to insure that the solution will not attack or errode the parts being cleaned. Water base detergent solutions shall be used for cleaning simple unchanneled parts. Detergent shall conform to MIL-D-16791, type I. Trichlorotrifluorethane shall be used for cleaning complex parts. Trichlorotrifluorethane *shall conform to MIL-C-81302.

5.1.1.1.7 Method g. Oxygen systems cleaning compound. Components shall be rinsed or flushed with cleaning compound conforming to MIL-C-81302. After cleaning, components shall be dried with filtered oil-free air or nitrogen.

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5.1.1.2 Parts that do not contact high-purity oxygen shall be cleaned and dried in accordance with the applicable processes and procedures of MIL-P-116 or as specified herein.

5.1.1.3 Packaging. Small components that contact high-purity oxygen shall be packaged in polyethylene bags conforming to MIL-B-117; or fabricated with polyethylene film conforming to L-P-378. Polyethylene film shall not be less than 6 mils in thickness. Prior to inserting into the bags, sharp surfaces and other irregular surfaces of the part shall be cushioned as required to prevent rupture or other damage to the bag. Folded pads, thick wraps, or twisted lengths used as cushioning shall conform to the same specifications as the bags. The cushioning shall be secured in place with vinyl or nylon cord. Prior to heat sealing, the bags shall be purged with filtered, oil-free, dry nitrogen. Large components which contact high-purity oxygen shall be purged with filtered, oil-free, dry nitrogen; and all openings sealed with blind flanges, expansion plugs, metal screw plugs, metal caps, polyethylene caps, or polyethylene film conforming to L-P-378 or equal. Sealing shall be accomplished in a manner to provide a positive seal which will prevent entrance of harmful foreign matter. Large or small components that contact high-purity oxygen that cannot be packaged as specified herein, shall be preserved per Method II in accordance with MIL-P-116.

* 5.1.1.4 Instrument panels. The glass fronts of instruments, instrument panels and dial glasses shall be protected by covering with neutral barrier material conforming to MIL-P-130 or MIL-P-17667. Glasses shall be bridged with pieces of fitted plywood or other similar protective material secured in place with tape conforming to PPP-T-60 or PPP-T-97.

5.1.1.5 Purity test sets. Purity test set components shall be individually wrapped with barrier material conforming to MIL-P-130 or MIL-P-17667 and cushioned with cushioning material conforming to PPP-C-843, type II, class A. Glassware shall be individually packaged in interior containers conforming to PPP-B-566, PPP-B-636, PPP-B-665 or PPP-B-676.

5.1.1.6 Electric motors and generators. Motors and generators shall be preserved Method II in accordance with MIL-P-116.

5.1.1.7 Push botton remote control stations. Push button remote control stations shall be packaged Method II in accordance with MIL-P-116.

* 5.1.1.8 Electric cables. Electric cables shall be coiled to the minimum safe dimensions and secured in not less than 3 places with tape conforming to PPP-T-97. Exposed ends of wires and sockets shall be sealed with plugs or caps conforming to MIL-C-5501 or sealed with tape conforming to PPP-T-60 class 1.

* 5.1.1.9 Pulleys. All unpainted pulleys shall be coated with compound conforming to TT-P-664, prior to belt installation. Belts shall be placed in operating position, whenever practicable, and tension completely removed. Removed belts shall be wrapped individually for single belt drives, or in sets for multiple drives with greaseproof barrier material conforming to MIL-B-121, grade A and sealed with heat sealer or pressure sensitive tape conforming to PPP-T-60. Packaged belts shall be secured to the equipment near pulleys on which they will be used. Waterproof tags, conforming to UU-T-81, shall be attached indicating the bolts proper application.

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5.1.1.10 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40.

5.1.1.11 Air compressors. The air compressors shall be preserved and packaged in accordance with MIL-C-3600.

5.1.1.12 Refrigeration and cooling equipment. Refrigeration and cooling equipment shall be preserved and packaged in accordance with MIL-R-3593.

5.1.1.13 Valves. Valves, except valves that contact high-purity oxygen, shall be preserved and packaged in accordance with MIL-V-3.

* 5.1.1.14 Piping and tubing. Piping and tubing of small diameter shall have the ends capped with plastic caps or plugs conforming to MIL-C-5501. Desiccant shall be placed in all piping and tubing that contact high-purity oxygen, when the diameter is of sufficient size to admit fractional quantities. The ends of large diameter piping and tubing which contact high-purity oxygen shall be sealed with gasketed cover plates, metal covers, or blind flanges. As an alternate, sealing can be accomplished by sealing the ends with barrier material conforming to MIL-B-131, class 1, secured with tape conforming to PPP-T-60. The taped edges shall be oversprayed with strippable coating in accordance with MIL-C-23760. Steel piping and tubing, that do not contact high-purity oxygen, shall be preserved and packaged in accordance with MIL-STD-163. Aluminum piping and tubing that do not contact high-purity oxygen shall be preserved and packaged in accordance with MIL-STD-649.

5.1.1.15 Intermediate packaging. Boxed items under 15 cubic inches and all bagged or wrapped items shall be packed in intermediate containers conforming to PP-B-636, type CF, class weather-resistant. Gross weight of intermediate containers shall not exceed 10 pounds or 750 cubic inches. Closure and sealing shall be in accordance with PPP-B-636 and appendix thereto.

5.1.2 Level C. Items which contact high purity oxygen shall be preserved and packaged the same manner as specified for Level A. Items that do not contact high-purity oxygen shall be preserved and packaged in accordance with the contractor's commercial practice.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified, components of the generating charging plant, preserved and packaged as specified in 5.1, shall be packed in containers conforming to PPP-B-601, overseas type, PPP-B-621, Class 2, MIL-C-104, MIL-C-3774, MIL-C-9897, or MIL-C-52950. The gross weight of each pack shall not exceed authorized weights specified in the containers specification. Containers shall be closed and strapped in accordance with the applicable container specification and appendix thereto.

5.2.2 Level B. Unless otherwise specified, components of the generating charging plants, preserved and packaged as specified in 5.1, shall be packed in containers conforming to PPP-B-601, domestic type, PPP-B-621, Class 1, PPP-B-636, Type SF, Class weather-resistant, PPP-B-640, Class 2, MIL-C-104, MIL-C-3774, MIL-C-9897, or MIL-C-52950. The gross weight of each pack shall not exceed weights specified in the container specification. Containers shall be closed and strapped in accordance with the applicable container specification and appendix thereto.

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5.2.3 Level C. Components of the generating charging plant, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules, or regulations of other carriers applicable to the mode of transportation.

5.3 Physical protection. Contents of shipping containers shall be anchored, blocked, braced, and cushioned in accordance with MIL-STD-1186. Testing: Free-fall drop test, edge-wise drop test, corner-wise drop test, and guided impact tests, shall be in accordance with appendix A, MIL-STD-1186. The tests are not required when Level C packing is specified.

5.4 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. In addition, to facilitate reassembly, parts that are removed, and the mating parts on the basic assembly, shall be matchmarked with lusterless white enamel, overcoated with water-resistant clear spar varnish conforming to TT-V-121. Small parts that cannot be matchmarked as specified above, shall be matchmarked with waterproof tags onforming to UU-T-81, class 2.

6. NOTES

6.1 Intended use. The liquid oxygen generating and charging plant described herein is intended for use in producing high purity oxygen and nitrogen for support of aircraft servicing requirements where building facilities suitable for erection and operation are available.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Level of cleaning, preservation, packaging, and packing required (see 5.1 and 5.2).

c. Location of first article testing (see 4.4).

6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing Activity:
Air Force - 82

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-83750B	Generating and Charging Plant, 5 Ton Per Day Capacity, Skid Mounted.

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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